Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

Forward deployed AI engineer specializing in **production LangGraph agents, RAG pipelines, and multi-provider LLM systems** - built AgentProbe as a public open-source tool with external API, documentation, and real user adoption. Designed the full agent architecture from scratch: Orchestrator (plan), RAG Tracer (execute + observe), Auto Fix (react + recover) - the same plan-execute-react loop enterprise AI deployments require. 8+ years production Python; API and data pipeline integration at scale. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Demo: *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

AI Agent Architecture | LangChain / LangGraph | RAG End-to-End | Function Calling & Tool Integration | Multi-Provider LLM Routing | Enterprise API Integration | Customer-Facing Technical Delivery | Python / FastAPI | AWS / Docker / CI/CD | LLM Evaluation & Observability | Cross-Functional Delivery | Production ML Monitoring

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Built **20+ production API integrations** (webhooks, REST, data pipelines) across external systems - PostgreSQL, AWS S3/Glue, microservices - processing **1M+ records** with automated quality checks; same integration pattern as enterprise AI agent deployments.
- Improved production query performance by **30%**; maintained **99.5% uptime** for 6+ months via TDD/BDD - zero security incidents; PHI-compliant data migration to AWS with full audit trails.
- Delivered **20+ full-stack modules** across **120+ Agile sprints** in cross-functional Product/Engineering/QA teams for **100K+ monthly active users**; cut microservices execution time by **50%** via Docker containerization.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Translated complex technical concepts for 30+ non-technical students; gathered feedback, iterated on delivery, adapted explanations to different knowledge levels - the same customer-facing technical communication FDE roles require.

PROJECTS

AgentProbe: Production AI Agent Platform OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · CI/CD

End-to-end AI agent platform demonstrating the full stack Cresta's FDE role builds and deploys. **Agent architecture**: 3 LangGraph agents with plan-execute-react loop - Orchestrator (plans multi-step runs), RAG Tracer (executes retrieval + function calls, observes quality), Auto Fix (reacts to failures, selects recovery strategy). **Function calling**: custom tool implementations dynamically selected by agents based on failure attribution - retrieval swap, prompt rewrite, provider switch. **RAG end-to-end**: ChromaDB vector store, embedding pipeline, retrieval scoring. **Customer-facing deployment**: 15-endpoint FastAPI backend, React dashboard, full documentation, public GitHub - designed for external adoption, not just personal use. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings · NumPy · Pandas

Enterprise-grade integration: NLP pipeline mapping natural language queries to structured FHIR APIs via LLMs, BERT, and embeddings. HIPAA-compliant RBAC. API + webhook integration with healthcare data systems. **Reduced retrieval time ~60%**.
github.com/venkatalakshmiparimala/AI_on_FHIR

CrashClarity: Predictive ML Pipeline on 100K+ Records

Python · R · Random Forest · Logistic Regression · Feature Engineering · Statistical Analysis · GitHub Pages

Full ML lifecycle: feature engineering → model comparison → 85%+ accuracy → interactive dashboard for non-technical stakeholders. Demonstrates data-driven decision-making from raw data to deployed insight.

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · ML systems, statistical modeling, experimental design, data pipelines at scale

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

AI & Agents: LangChain, LangGraph, Multi-Provider LLM (Anthropic, OpenAI, Groq, Ollama), RAG Pipelines, ChromaDB, Function Calling, Custom Tools, Prompt Engineering, LLM Evaluation, FastAPI

ML & Data: scikit-learn, PyTorch, TensorFlow, Random Forest, Feature Engineering, NumPy, Pandas, BERT, spaCy, Hugging Face

Cloud & Infra: AWS (S3, EC2, RDS, Glue), Azure, GCP-compatible, PostgreSQL, MySQL, NoSQL, Docker, CI/CD (GitHub Actions), RabbitMQ, Microservices

Integration: REST APIs, Webhooks, Data Pipelines, TDD/BDD, Git, JIRA, Postman, Agile/Scrum